

DINESH VIJAYAKUMAR

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Professional Summary

Software Engineer with 12+ years of experience in back-end, building and maintaining reliable, scalable distributed systems in production. Domain expert in E-commerce, Supply Chain and Logistics. Experienced in designing, developing and troubleshooting containerized services in AWS Cloud. Passionate about Big Data Technologies and Developer Tooling with familiarity in Machine Learning.

Work Experience

Amazon, India | Sr. SDE | 05/2021 – Present

- Delivered ICQA (aka Audit for SOX Compliance) capability and a suite of problem solve tools to enable full-launch of warehouse management software (WMS) in multiple distribution centers across North America achieving a productivity gain of **~93%** with the new audit feature.
- Designed and developed a cross-WMS compatible software solution for grading produce received in distribution centers against USDA food safety guidelines.
- Implemented performance optimizations in a critical backend service to result in **50%** fleet size reduction worldwide.
- Facilitated over **200+** interviews as a Bar Raiser for Interviewing and conducted org-wide operational reviews.

Amazon, US | Sr. SDE | 10/2017 – 04/2021

- Drove the effort to add Bitbucket support in AWS Code Guru Reviewer in April 2020 and GA launch the service in May 2020 with Github Enterprise support.
- Led the design and development of Automated Surge Pricing for Amazon Flex delivery program. It has resulted in savings of ~100 man-hours per week and enabled us to sustain higher rates of filling labor supply (>97%) in US alone while lowering surge cost per hour (CPH) to **~11 cents** on average.
- Spearheaded an org-wide tech initiative “Last Mile Data Lake” to develop AWS S3-based data lake as a cost-effective and centralized solution for analysis and reporting as opposed to the fragmented solution using Amazon Redshift clusters managed by individual teams. This was estimated to save **~\$700K** in data warehousing costs annually.
- Promoted to Senior SDE for architecting the design of an automated solution to build and deploy machine learning models at scale. This involved building large scale software components for data ingestion, storage, analytics, feature engineering, model training and evaluation.

Skills & Competencies

- Java
- Kotlin
- Object-Oriented design and programming (OOP)
- Software Development Life-Cycle (SDLC)
- Cloud Technologies (AWS)
- Service-Oriented Architecture
- Automated Testing (Unit/Functional/Integration)
- Software Debugging & Troubleshooting
- Scripting and Automation
- Code Refactoring
- Version Control (Git)
- API Design & Development (REST, RPC, GraphQL)
- Docker & Containerization
- Algorithms & Data Structures
- System Architecture Design & Implementation
- Continuous Integration/Continuous Delivery
- Technical Writing and Documentation
- Software Development Methodologies
- Database Design & Modeling
- Big Data Technologies

Education

B.E in Computer Science & Engineering, Graduation Year 2011

Government College of Technology, Coimbatore, India.

GPA: 8.98/10 (Distinction)

Gold Medalist, TCS Best Student & GCT Alumni Merit Scholarship Awardee

Certifications & Courses

2009 | Oracle Certified Java Programmer – 5.0 | Certificate ID – SR5715191

2021 | Kotlin for Java Developers | Coursera

Awards & Accolades

2011 | Srujan Contest Runners | Amazon (“*Product Recommendations through Social Network*”)

2017 | Last Mile Defect Destroyer | Amazon

Amazon, US | SDE | 07/2014 – 09/2017

- Redesigned the API and Data model of Amazon Flex Payments systems enabling revamp of earnings view experience in the mobile app used by Delivery Partners (DPs). This effort reduced the user facing API latencies by **66% (TP99)**.
- Automated the invoicing and disbursement workflow for paying over **millions of dollars** every month to Delivery Service Providers (DSPs) who partner with Amazon for providing last-mile delivery services.
- Automated OS patching of EC2 servers running applications to avoid **~10%** of software compliance issues reducing operational burden.
- Recognized with “**Last Mile Defect Destroyer 2017 Award**” for significant contributions to improving code quality as well as developing a tool for troubleshooting application failures.
- Senpai Trainer and Application Security Certifier since 2017.
- Invited as an **honorary guest** (1 out of 5) by Service Frameworks team for feedback to proposed enhancements to SOA framework widely used at Amazon, on account of useful contributions in educating the larger SDE Community about the framework capabilities.
- Presented the topic “Growing from SDE I to SDE II” as part of Amazon Insider Training in March 2017 to an audience of **~50** budding SDEs aspiring to grow their career at Amazon.
- Designed and Implemented features to ensure idempotency for EBS Volume APIs to eliminate the top root cause of EC2 instance launch failures arising from transient request failures.
- Worked on the release of Throughput Optimized HDD (st1) and Cold HDD (sc1) volume types at EBS for offering AWS customers predictable performance for throughput-intensive and big data workloads with large data sets, large input/output (I/O) block sizes, and sequential I/O patterns.
- Developed Amazon SWF-based workflow solution that isolates hundreds of EBS Storage Servers failing to pass the diagnostic tests to make sure EBS Volumes are always available despite server failures.
- Added features to Gossip Protocol-based daemon, a highly scalable distributed system deployed throughout AWS for endpoint discovery and failure detection and this was critical to maintaining high availability of this system (**99.99%**).

Amazon, India | SDE | 07/2011 – 07-2014

- Developed an external portal enabling Amazon’s 3P carriers to report on shipments damaged/mis-sorted in transit and the backend systems to issue concessions to impacted customers. This meant the SLA for reporting and reacting to such exceptional events improved from ~2 days on average to a matter of few seconds.
- Developed a highly available data lookup service that helped multiple development teams across Transportation to query for aggregated views of shipment level details. This service supported **>300 TPS** at its peak capacity during the holiday season of 2012.
- SDE Bootcamp trainer and University Relations Panelist at Amazon, India conducting over 30+ sessions for new hires joining Amazon.